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Electric work machine

Abstract

An electric work machine includes a first and second motors; a first and second switches disposed on corresponding current conduction path from a direct-current power source to each motor; a first and second carrier frequency setting devices; and a controller. The first and second carrier frequency setting devices respectively set a first and second carrier frequencies of a time when PWM controlling an electric power supply to the first and second motors. The first and second carrier frequencies are different from each other. The controller drives the first switch or the second switch by using a PWM signal generated at the first or second carrier frequency and PWM controls an electric power supply to the first or second motor.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims the benefit of Japanese Patent Application No. 2021-168021 filed Oct.

13, 2021 in the Japan Patent Office, the entire disclosure of which is incorporated herein by reference.

BACKGROUND

(2) The present disclosure relates to an electric work machine.

(3) International Patent Application Publication No. 2019/097683 (hereinafter, Patent Document 1) discloses a lawn mower that includes an operation motor, a travel motor, and a controller. In this lawn mower, the operation motor is a motor for driving a cutting blade, and the travel motor is a motor for causing the lawn mower to travel (self-travel). The controller includes a computer in which a processor executes a program stored in a memory to control a rotational speed of the operation motor and a rotational speed of the travel motor.

SUMMARY

(4) A rotational speed of a motor is controlled by turning on and off a switching element disposed on a current conduction path to the motor by using a pulse-width modulated (PWM) control signal (PWM signal). If a carrier frequency, which determines a cycle of PWM control, is high, the operation motor sometimes brings about a failure of switching element caused by overheating due to switching losses. It is thus required to reduce the carrier frequency to reduce the number of times of switching. Meanwhile, it is required to increase the carrier frequency of the travel motor to stabilize the torque (in other words, the rotational speed of the motor) since the lawn mower travels unstably to decrease its usability when torque fluctuation occurs.

(5) However, the lawn mower disclosed in Patent Document 1 includes one controller to control both the operation motor and the travel motor. As both motors are controlled by a common controller, the carrier frequency of a time when PWM controlling each motor becomes the same. Accordingly, the lawn mower disclosed in Patent Document 1 has a problem that it is not possible to set an appropriate carrier frequency for each motor to control the rotational speed of each motor.

(6) In one aspect of the present disclosure, it is desirable that, in an electric work machine having two motors, a common controller for the two motors can control an electric power supply to each motor by using PWM signals having different carrier frequencies.

Means for Solving the Problems

(7) An electric work machine in one aspect of the present disclosure includes a first motor, a second motor, a first current conduction path, a second current conduction path, a first switch, a second switch, a first carrier frequency setting device, a second carrier frequency setting device, and a controller.

(8) The first motor is configured to drive a first drive object. The second motor is configured to drive a second drive object. In other words, the first motor and the second motor drive different drive objects.

(9) The first current conduction path is configured to couple a direct-current power source to the first motor. The second current conduction path is configured to couple the direct-current power source to the second motor. A first switch is disposed on the first current conduction path. A second switch is disposed on the second current conduction path.

(10) The first carrier frequency setting device is configured to set a first carrier frequency of a time when PWM controlling an electric power supply to the first motor by turning on and off the first switch. The second carrier frequency setting device is configured to set a second carrier frequency of a time when PWM controlling an electric power supply to the second motor by turning on and off the second switch such that the second carrier frequency is different from the first carrier frequency.

(11) The controller is configured to control an electric power supply to the first motor by generating a first PWM signal at the first carrier frequency and delivering the first PWM signal to the first switch in response to an input of a first motor drive command. The controller is also configured to control an electric power supply to the second motor by generating a second PWM signal at the second carrier frequency and delivering the second PWM signal to the second switch in response to

an input of a second motor drive command.

(12) As described above, in the aforementioned electric work machine, the electric power supplies to the first motor and the second motor are PWM controlled by the first PWM signal and the second PWM signal that are respectively generated at the first carrier frequency and the second carrier frequency having different frequencies.

(13) Accordingly, in the aforementioned electric work machine, even though the first motor and the second motor are driven by a common controller, the carrier frequency of a time when PWM controlling the electric power supply to the first motor and the second motor can be set individually for each motor. Therefore, the electric power supply to the first motor and the second motor can be PWM controlled individually at the carrier frequency suitable for each motor.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) An example embodiment of the present disclosure will be described hereinafter by way of example with reference to the accompanying drawings, in which:

(2) FIG. 1 is a perspective view showing an external appearance of a lawn mower in an embodiment;

(3) FIG. 2 is a block diagram showing an electrical configuration of the lawn mower;

(4) FIG. 3 is a time chart showing control operation of a blade motor and a self-travel motor when the blade motor and the self-travel motor are driven;

(5) FIG. 4 is a flow chart showing a control process executed by a controller;

(6) FIG. 5 is a flow chart showing a motor drive process shown in FIG. 4;

(7) FIG. 6 is a flow chart showing a blade motor control process shown in FIG. 5;

(8) FIG. 7 is a flow chart showing a self-travel motor control process shown in FIG. 5;

(9) FIG. 8 is a flow chart showing a self-travelling brake control process shown in FIGS. 4, 5, 7;

(10) FIG. 9 is a flow chart showing a blade brake control process shown in FIGS. 4, 5, 6; and

(11) FIG. 10 is a time chart showing operation of the blade motor when break control is executed.

DETAILED DESCRIPTION OF EXEMPLARY EMBODIMENTS

Embodiment

(12) An electric work machine of the present embodiment may include a first motor configured to drive a first drive object. The first motor may include a first current conduction path configured to couple a direct-current power source to the first motor. A first switch may be disposed on the first current conduction path.

(13) In addition or alternatively, the electric work machine may include a second motor configured to drive a second drive object. The second motor may include a second current conduction path configured to couple the direct-current power source to the second motor. A second switch may be disposed on the second current conduction path.

(14) In addition or alternatively, the electric work machine may include a first carrier frequency setting device configured to set a first carrier frequency of a time when PWM controlling an electric power supply to the first motor by turning on and off the first switch. The electric work machine may also include a second carrier frequency setting device configured to set a second carrier frequency of a time when PWM controlling an electric power supply to the second motor by turning on and off the second switch such that the second carrier frequency is different from the first carrier frequency.

(15) In addition or alternatively, the electric work machine may include a controller that controls the electric power supply to the first motor by generating a first PWM signal at the first carrier frequency and delivering the generated first PWM signal to the first switch in response to an input of a first motor drive command.

(16) This controller may also be configured to control the electric power supply to the second motor by generating a second PWM signal at the second carrier frequency and delivering the generated second PWM signal to the second switch in response to an input of a second motor drive command.

(17) An electric work machine including the first motor, the second motor, the first current conduction path, the second current conduction path, the first switch, the second switch, the first carrier frequency setting device, the second carrier frequency setting device, and the controller described above may set the first carrier frequency for PWM controlling the electric power supply to the first motor to a carrier frequency suitable to control the first motor and may set the second carrier frequency for PWM controlling the electric power supply to the second motor to a carrier frequency suitable to control the second motor.

(18) For example, in a case where a load applied from the first drive object to the first motor is large and thus the electric power supply to the first motor is large, the first carrier frequency may be set to a frequency that is lower than the second carrier frequency. By thus setting the first carrier frequency, switching losses generated in the first switch can be reduced, which can inhibit the first switch from going into an overheated state.

(19) In this case, it is not necessary to increase a dielectric withstanding voltage of the first switch to protect the first switch from failure caused by overheating. This makes it possible to select a switching element having a low dielectric withstanding voltage as the first switch. Accordingly, the cost of the electric work machine can be reduced.

(20) In this case, the second carrier frequency may be set higher than the first carrier frequency. This makes it possible to reduce torque fluctuation of the second motor and drive the second drive object stably at a desired speed.

(21) In addition or alternatively, in the electric work machine, the first carrier frequency may be set lower than the second carrier frequency. The first drive object may be a blade. The first motor may be configured to rotate the blade. The second drive object may be a wheel that enables the electric work machine to travel. The second motor may be configured to rotate the wheel.

(22) In the first motor, a load generated by driving the blade increases when operating the electric work machine. The first switch is therefore likely to generate heat because of switching losses caused by being periodically turned on and off by PWM control. However, such heat generation can be inhibited by setting the first carrier frequency lower than the second carrier frequency as mentioned above.

(23) In a case where the second motor has a low carrier frequency, torque fluctuation (in other words, rotational fluctuation) is likely to occur in the second motor because of a load generated by driving the wheel. Such torque fluctuation can be inhibited by increasing the second carrier frequency. This stabilizes the travelling speed of the electric work machine when self-travelling and enables a reduction of uncomfortableness felt by the user due to speed fluctuation.

(24) In addition or alternatively, the first carrier frequency of the electric work machine may be set within a range between 100 Hz and 1 kHz; and the second carrier frequency may be set within a range between 8 kHz and 20 kHz.

(25) The electric work machine having the first carrier frequency or the second carrier frequency set as specified above can reduce a switching noise to be heard by the user when PWM controlling the electric power supply to the first motor and/or when PWM controlling the electric power supply to the second motor.

(26) Generally, in a case where the carrier frequency of the PWM control is set within a range between 1 kHz and 8 kHz, the switching noise generated by turning on and off the switches becomes an audible noise that can be heard by a person in the surroundings and may give uncomfortableness to the users and others.

(27) However, if the first carrier frequency or the second carrier frequency is set as specified above, it is possible to reduce the switching noise generated when PWM controlling the electric power

supply to the motors, which enables a reduction of uncomfortableness felt by the user.

(28) More preferably, the first carrier frequency may be set within a range between 300 Hz and 700 Hz; and the second carrier frequency may be set within a range between 8 kHz and 10 kHz.

(29) In addition or alternatively, the first switch and/or the second switch may be configured such that a turn-off time, during which the state of the switch changes from ON-state to OFF-state by a first PWM signal and/or a second PWM signal, is longer than a turn-on time, during which the state of the switch changes from OFF-state to ON-state.

(30) The electric work machine configured as described above can reduce a high-value peak voltage that is generated by energy accumulated in the first motor or in the second motor when the first switch or the second switch is turned off. As a result, it is possible to inhibit the first switch or the second switch from being deteriorated due to a high voltage that is generated when the first switch or the second switch is turned off.

(31) The aforementioned features may be combined in any ways as another embodiment. One or some of the aforementioned features may be deleted as yet another embodiment.

Specific Example Embodiment

(32) Hereinafter, example embodiments of the present disclosure will be explained with reference to the drawings. In the present embodiment, a lawn mower **10** will be explained as an example of the electric work machine.

(33) <Overall Configuration of Lawn Mower>

(34) As shown in FIG. **1**, the lawn mower **10** of the present embodiment includes a vehicle body **11**. The lawn mower **10** also includes a battery connector **19** above the vehicle body **11**. The battery connector **19** includes a lid. The battery connector **19** is configured so that two or more battery packs **6** (see FIG. **2**) can be attached thereto or detached therefrom inside the lid. Each battery pack **6** include a rechargeable battery **8**. The battery **8** is one example of a direct-current power source in the present disclosure.

(35) The lawn mower **10** includes a grass collector **5**. The grass collector **5** is attached to the rear of the vehicle body **11**.

(36) The lawn mower **10** includes a handle **12**. The handle **12** has a U-shaped form and is attached to the vehicle body **11**. The handle **12** is configured to be held by the user. The user holds the handle **12**, walk behind the lawn mower **10**, and controls the lawn mower **10** via the handle **12**.

(37) The lawn mower **10** includes a first control lever **13** and a second control lever **14** for the user to control. These two control levers **13**, **14** have U-shaped forms to correspond to the shape of the handle **12**. The first control lever **13** is situated in front of the handle **12**. The second control lever **14** is situated in the rear of the handle **12**.

(38) The first control lever **13** is attached to the handle **12** so that the user can pull the first control lever **13** towards the handle **12** (in other words, towards the rear). The second control lever **14** is attached to the handle **12** so that the user can push the second control lever **14** towards the handle **12** (in other words, towards the front).

(39) The lawn mower **10** includes a cutting blade **18** (see FIG. **2**). The cutting blade **18** is disposed on a bottom surface of the vehicle body **11**. The lawn mower **10** includes a blade motor **31** (see FIG. **2**). The blade motor **31** is disposed inside the vehicle body **11**. A rotational shaft of the blade motor **31** is physically coupled to the cutting blade **18** directly or via a transmitting device.

(40) The blade motor **31** is a motor for driving the cutting blade **18** for mowing the lawn grasses. The cutting blade **18** rotates in response to receiving a driving force from the blade motor **31**. The lawn grasses mown by the rotation of the cutting blade **18** is collected inside the grass collector **5**.

(41) The lawn mower **10** includes a pair of right and left drive wheels **16**. The pair of drive wheels **16** is disposed in the rear of the vehicle body **11**. The lawn mower **10** includes a pair of right and left driven wheels **15**. The pair of driven wheels **15** is disposed in the front of the vehicle body **11**.

(42) The lawn mower **10** includes a self-travel motor **32** inside the vehicle body **11** (see FIG. **2**). A rotational shaft of the self-travel motor **32** is physically coupled to the pair of drive wheels **16** via a

transmitting device. The self-travel motor **32** is a motor for driving the pair of drive wheels **16**. The pair of drive wheels **16** rotates in response to receiving a drive force from the self-travel motor **32** and cause the lawn mower **10** to self-travel towards the front.

(43) The blade motor **31** corresponds to one example of the first motor in the present disclosure. The cutting blade **18** is a blade in the present disclosure and corresponds to one example of the first drive object. The self-travel motor **32** corresponds to one example of the second motor in the present disclosure. The drive wheels **16** are one example of the second drive object in the present disclosure. In the present embodiment, the blade motor **31** and the self-travel motor **32** may be brushed motors.

(44) Among the two control levers **13**, **14**, the first control lever **13** is a control lever for driving or stopping the blade motor **31**; and the second control lever **14** is a control lever for driving or stopping the self-travel motor **32**.

(45) The first control lever **13** includes a blade switch **21** (see FIG. 2) as a switch for driving the blade motor **31**. The blade switch **21** turns to ON-state when the first control lever **13** is pulled towards the handle **12** (in other words, towards the rear).

(46) The second control lever **14** includes a self-travel switch **22** (see FIG. 2) as a switch for driving the self-travel motor **32**. The self-travel switch **22** turns to ON-state when the second control lever **14** is pushed towards the handle **12** (in other words, towards the front).

(47) The first control lever **13** and the second control lever **14** are attached to the handle **12** such that they return to their original position when released by the user. Accordingly, when the first control lever **13** is pulled, the blade switch **21** returns to OFF-state if the user releases the first control lever **13**. Similarly, when the second control lever **14** is pushed, the self-travel switch **22** returns to OFF-state if the user releases the second control lever **14**.

(48) The lawn mower **10** includes an operation device **17**. The operation device **17** is attached to the handle **12**. The operation device **17** includes a main power switch **20**. The main power switch **20** is a tactile switch that turns to ON-state only when the user is manipulating (pressing) the switch.

(49) The operation device **17** also includes a display **26** that displays state of the lawn mower **10**. The display **26** includes LEDs for various different displays. The LEDs of the display **26** include, for example, a main power LED that is turned on when electric power is supplied to a controller **30** in response to a manipulation of the main power switch **20** as shown in FIG. 2. The LEDs of the display **26** also include a remaining energy display LED that shows electric energy accumulated in the battery **8** (hereinafter referred to as remaining energy), and a malfunction display LED that is turned on when the driving system of the blade motor **31** and/or the self-travel motor **32** is malfunctioning.

(50) The operation device **17** also includes a speed adjusting dial **24** for adjusting travelling speeds of the aforementioned blade switch **21**, the self-travel switch **22**, and the lawn mower **10**. As shown in FIG. 2, the speed adjusting dial **24** includes a variable resistor, a resistance value of which varies depending on the position of the speed adjusting dial **24** controlled by the user. The controller **30** is configured to transmit (generate) a command with respect to the travelling speed in accordance with the resistance value.

(51) <Configuration of Controller>

(52) The controller **30** drives the blade motor **31** and the self-travel motor **32** and controls the display of various different LEDs arranged on the display **26**. The controller **30** is disposed on a single circuit board and stored inside the vehicle body **11**. Hereinafter, an electric configuration of the controller **30** will be explained with reference to FIG. 2.

(53) As shown in FIG. 2, the controller **30** is electrically coupled to the battery **8**, the blade motor **31**, and the self-travel motor **32**. The controller **30** includes a first current conduction path **35**, a second current conduction path **36**, and a third current conduction path **34**. The first current conduction path **35** and the third current conduction path **34** couple the battery **8** to the blade motor

31. The third current conduction path **34** and the second current conduction path **36** couple the battery **8** to the self-travel motor **32**. This enables electric power supplies from the battery **8** to the blade motor **31** and to the self-travel motor **32**.

(54) The first current conduction path **35** and the second current conduction path **36** are respectively dedicated for the blade motor **31** and the self-travel motor **32**. The first current conduction path **35** is coupled to two terminals of the blade motor **31**, and the second current conduction path **36** is coupled to two terminals of the self-travel motor **32**. A first terminal of these two terminals of the blade motor **31** is coupled to a positive side of the battery **8**, and a second terminal of these two terminals is coupled to a negative side of the battery **8** (which is the ground in FIG. 2). The two terminals of the self-travel motor **32** are arranged likewise. The third current conduction path **34** is a common current conduction path for the blade motor **31** and the self-travel motor **32** that couples the positive side terminal battery **8** to the first current conduction path **35** and the second current conduction path **36**.

(55) The third current conduction path **34** has a third switching element **40** arranged thereon. The third switching element **40** completes and interrupts the third current conduction path **34**. The third switching element **40** is an re-channel type metal-oxide semiconductor field-effect transistor (hereinafter, referred to as MOSFET). Accordingly, the third switching element **40** completes the third current conduction path **34** by turning to ON-state in response to an input of a high-level drive signal into its gate from the control circuit **50**.

(56) On a part of the first current conduction path **35** that couples the terminal of the blade motor **31** to the negative side of the battery **8**, a first switching element **41** that completes and interrupts the first current conduction path **35** is disposed. On a part of the second current conduction path **36** that couples the terminal of the self-travel motor **32** to the negative side of the battery **8**, a second switching element **42** that completes and interrupts the second current conduction path **36** is disposed.

(57) The first and the second switching elements **41**, **42** are n-channel type MOSFET like the third switching element **40**. Accordingly, the first and the second switching elements **41**, **42** respectively complete the first current conduction path **35** and the second current conduction path **36** by turning to ON-state in response to an input of a high-level drive signal into their gates from the control circuit **50**.

(58) When driving the blade motor **31** or the self-travel motor **32**, the control circuit **50** generates a PWM signal having a specific duty ratio at a carrier frequency set for each of the blade motor **31** and the self-travel motor **32**. The control circuit **50** then delivers thus generated PWM signal to the first and the second switching elements **41**, **42**, which are respectively corresponding to the blade motor **31** and the self-travel motor **32**, to turn on and off the first and the second switching elements **41**, **42**. As a consequence, electric power supply to the blade motor **31** and the self-travel motor **32** is PWM controlled.

(59) The first switching element **41** corresponds to one example of the first switch in the present disclosure. The PWM signal delivered from the control circuit **50** to the first switching element **41** corresponds to one example of the first PWM signal in the present disclosure. The second switching element **42** corresponds to one example of the second switch in the present disclosure. The PWM signal delivered from the control circuit **50** to the second switching element **42** corresponds to one example of the second PWM signal in the present disclosure.

(60) A resistor **R11** is arranged on a part of the first current conduction path **35** between the first switching element **41** and the negative side of the battery **8**. A resistor **R21** is arranged on a part of the second current conduction path **36** between the second switching element **42** and the negative side of the battery **8**. These resistors **R11**, **R21** are current detection resistors for detecting motor current flowing through the blade motor **31** and the self-travel motor **32**. The resistor **R11** includes a first current detection circuit **43** for detecting the motor current that flowed to the blade motor **31** from a voltage across the resistor **R11**. The resistor **R21** includes a second current detection circuit

44 for detecting the motor current that flowed to the self-travel motor **32** from a voltage across the resistor **R21**.

(61) Detection signals from the first current detection circuit **43** and the second current detection circuit **44** are input to the control circuit **50**. The control circuit **50** monitors the motor current flowed to the blade motor **31** when the blade motor **31** is driven based on the detection signal delivered from the first current detection circuit **43**. The control circuit **50** also monitors the motor current flowed to the self-travel motor **32** when the self-travel motor **32** is driven based on the detection signal delivered from the second current detection circuit **44**.

(62) If the motor current monitored by the control circuit **50** exceeds a preset electric current threshold, then the control circuit **50** determines that a driving system of the blade motor **31** or the self-travel motor **32** is malfunctioning and turns the third switching element **40** from ON-state to OFF-state to interrupt the third current conduction path **34**. As a consequence, the blade motor **31** and the self-travel motor **32** are protected from overcurrent.

(63) The first current conduction path **35** includes a fourth switching element **45** arranged in parallel with the blade motor **31**. The second current conduction path **36** includes a fifth switching element **46** arranged in parallel with the self-travel motor **32**.

(64) The fourth and the fifth switching elements **45**, **46** are n-channel type MOSFET like the first to the third switching elements **40**, **41**, **42**. Accordingly, the fourth and the fifth switching elements **45**, **46** respectively short circuits both ends of the blade motor **31** and the self-travel motor **32** by turning to ON-state in response to an input of a high-level brake signal into their gates from the control circuit **50**.

(65) Consequently, if the first switching element **41** is turned to OFF-state and the fourth switching element **45** is turned to ON-state when the blade motor **31** is rotating, a brake current flows through the fourth switching element **45** which causes generation of a damping torque in the blade motor **31**. If the second switching element **42** is turned to OFF-state and the fifth switching element **46** is turned to ON-state when the self-travel motor **32** is rotating, a brake current flows through the fifth switching element **46** which causes generation of the damping torque in the self-travel motor **32**. In other words, the fourth and the fifth switching elements **45**, **46** are short circuit brake switches that generate a braking force in the blade motor **31** and the self-travel motor **32** to stop their rotation.

(66) A drive switch **21A** of the blade motor **31** is disposed on a part of the first current conduction path **35** between the blade motor **31** and the fourth switching element **45**, and the positive side of the battery **8**. The drive switch **21A** is turned on and off in conjunction with the blade switch **21**. A drive switch **22A** of the self-travel motor **32** is disposed on a part of the second current conduction path **36** between the self-travel motor **32** and the fifth switching element **46**, and the positive side of the battery **8**. The drive switch **22A** is turned on and off in conjunction with the self-travel switch **22**.

(67) The drive switch **21A** completes the current conduction path **35** when the drive command of the blade motor **31** is input in response to the blade switch **21** being turned to ON-state. The drive switch **22A** completes the current conduction path **36** when the drive command of the self-travel motor **32** is input in response to the self-travel switch **22** being turned to ON-state.

(68) Consequently, if the first switching element **41** malfunctioned due to short circuit, the drive switch **21A** is turned off in conjunction with the blade switch **21** being turned off to interrupt the first current conduction path **35**; and if the second switching element **42** malfunctioned due to short circuit, the drive switch **22A** is turned off in conjunction with the self-travel switch **22** being turned off to interrupt the second current conduction path **36**. Therefore, even when the first and the second switching elements **41**, **42** malfunctioned due to short circuit, the user can still stop the drive of the blade motor **31** or the self-travel motor **32** by respectively turning the blade switch **21** or the self-travel switch **22** to OFF-state.

(69) A first turn-off-delay circuit **47** is disposed on an output path of the PWM signal from the control circuit **50** to the first switching element **41**. A second turn-off-delay circuit **48** is disposed

on an output path of the PWM signal from the control circuit **50** to the second switching element **42**. The first and the second turn-off-delay circuits **47**, **48** makes a turn-off time, in which the first and the second switching elements **41**, **42** are turned from ON-state to OFF-state in response to the PWM signal, longer than a turn-on time, in which the first and the second switching elements **41**, **42** are turned from OFF-state to ON-state in response to the PWM signal.

(70) Specifically, the first turn-off-delay circuit **47** includes a resistor **R12** disposed on the output path of the PWM signal from the control circuit **50** to the first switching element **41**, and a series circuit including a resistor **R13** and a diode **D11** coupled in parallel with the resistor **R12**. The diode **D11** is arranged such that a direction of electric current that flows from the control circuit **50** to the gate of the first switching element **41** is directed forward when the PWM signal rises from a low level to a high level.

(71) Therefore, the electric current that flows into the gate of the first switching element **41** from the control circuit **50** when the PWM signal rises to a high level is greater than the electric current that flows out of the gate of the first switching element **41** to the controller **50** when the PWM signal is weakened to a low level. Accordingly, the turn-off time, in which the first switching element **41** turns from ON-state to OFF-state in response to the PWM signal being weakened to a low level, is longer than the turn-on time, in which the first switching element **41** turns from OFF-state to ON-state in response to the PWM signal rising to a high level.

(72) The second turn-off-delay circuit **48** includes a resistor **R22** disposed on the output path of the PWM signal from the control circuit **50** to the second switching element **42**, and a series circuit including a resistor **R23** and a diode **D21** coupled in parallel with the resistor **R22**. Similarly to the diode **11** of the first turn-off-delay circuit **47**, the diode **D21** is arranged such that a direction of electric current that flows from the control circuit **50** to the gate of the second switching element **42** is directed forward when the PWM signal rises. Accordingly, the turn-off time, in which the second switching element **42** turns from ON-state to OFF-state, is longer than the turn-off time, in which the second switching element **42** turns from OFF-state to ON-state.

(73) The reason that the turn-off time of the first and the second switching elements **41**, **42** is made longer than their turn-on time is to reduce the high-value peak voltage generated when the first and the second switching elements **41**, **42** are turned off.

(74) When the first and the second switching elements **41**, **42** are turned from ON-state to OFF-state, a high voltage is generated at the terminals of the blade motor **31** and the self-travel motor **32** close to the first and the second switching elements **41**, **42** by energy accumulated in the blade motor **31** and the self-travel motor **32**. If this high-value peak voltage exceeds the dielectric withstanding voltages of the first and the second switching elements **41**, **42**, the first and the second switching elements **41**, **42** are deteriorated and broken in some cases. In the present embodiment, this high-value peak voltage generated by energy accumulated in the blade motor **31** and the self-travel motor **32** is reduced by making the turn-off time of the first and the second switching elements **41**, **42** longer, which protects the first and the second switching elements **41**, **42** from high voltages.

(75) The controller **30** includes a capacitor **37** for stabilizing the voltage of the third current conduction path **34** that receives electric power supply from the battery **8**. The controller **30** also includes a power-supply circuit **38** that generates a power supply voltage **Vcc** in response to receiving electric power supply from the battery **8** via the third current conduction path **34**.

(76) The power supply voltage **Vcc** generated in the power-supply circuit **38** is a direct-current constant voltage that can drive the control circuit **50** and is supplied to the control circuit **50**. The power supply voltage **Vcc** is also applied to the main power switch **20**, the blade switch **21**, and the self-travel switch **22** respectively via resistors **R0**, **R1**, **R2**. The power supply voltage **Vcc** is also applied to the variable resistor of the speed adjusting dial **24**.

(77) A connection point between the main power switch **20** and the resistor **R0**, a connection point between the blade switch **21** and the resistor **R1**, and a connection point between the self-travel

switch **22** and the resistor **R2** are each coupled to the control circuit **50**. A slider of the variable resistor of the speed adjusting dial **24** is also coupled to the control circuit **50**. Therefore, the control circuit **50** can obtain ON-state and OFF-state of the main power switch **20**, the blade switch **21**, and the self-travel switch **22** as well as a command value (speed command value) of the travelling speed from the speed adjusting dial **24**.

(78) The connection point between the main power switch **20** and the resistor **R0** is also coupled to the power-supply circuit **38**. As shown in FIG. 3, the power-supply circuit **38** is activated or stopped every time the main power switch **20** is manipulated and turned to ON-state. In other words, since the main power switch **20** is turned to ON-state only when manipulated (pressed) by the user, the state of the power-supply circuit **38** is changed between ON-state and OFF-state every time the main power switch **20** is manipulated.

(79) The controller **30** includes a display circuit **58** that turns on or off the LEDs arranged on the display **26** for various different displays. The control circuit **50** controls displayed state of the LEDs on the display **26** via this display circuit **58**.

(80) <Functional Configuration of Control Circuit>

(81) The control circuit **50** is a micro control unit (MCU) including a CPU, a ROM, a RAM, and the like. The control circuit **50** achieves each function described in FIG. 2 by the CPU executing programs stored in the ROM.

(82) In other embodiments, the control circuit **50** may include a combination of electronic components, such as a discrete element, or an application specific integrated circuit (ASIC) in place of or in addition to the MCU. The control circuit **50** may include an application specific standard product (ASSP), a programmable logic device such as a field programmable gate array (FPGA), or a combination of the above.

(83) Functions of the control circuit **50** will be explained next.

(84) As shown in FIG. 2, the control circuit **50** includes functions of a speed command determining device **52**, an SW input determining device **54** (SW: switch), a display controller **56**, a clock generator **60**, a first divider **61**, a second divider **62**, a first PWM generator **65**, a second PWM generator **66**, and a motor drive controller **68**.

(85) The speed command determining device **52** obtains a voltage value, which is obtained by dividing the power supply voltage V_{cc} by the variable resistor of the speed adjusting dial **24**, from the slider of the variable resistor and determines a speed command value of the travelling speed.

(86) The SW input determining device **54** determines ON-state and OFF-state of the main power switch **20**, the blade switch **21**, and the self-travel switch **22** from electric potentials at the connection point between the main power switch **20** and the resistor **R0**, the connection point between the blade switch **21** and the resistor **R1**, and the connection point between the self-travel switch **22** and the resistor **R2**.

(87) When the electric potential of the aforementioned connection point is low, the SW input determining device **54** determines that the corresponding switch **20**, **21**, or **22** is in ON-state; when the electric potential of the aforementioned connection point is high, the SW input determining device **54** determines that the corresponding switch **20**, **21**, or **22** is in OFF-state.

(88) The clock generator **60** is a so-called oscillator that generates a reference clock for determining operation timing of the control circuit **50**.

(89) The first divider **61** sets a cycle of the first PWM signal, in other words, the first carrier frequency for PWM controlling electric power supply to the blade motor **31** by dividing the reference clock delivered from the clock generator **60**.

(90) The second divider **62** sets a cycle of the second PWM signal, in other words, the second carrier frequency for PWM controlling electric power supply to the self-travel motor **32** by dividing the reference clock delivered from the clock generator **60**.

(91) As shown in FIG. 3, when the blade switch **21** is in ON-state, the first PWM generator **65** generates the first PWM signal to PWM control the electric power supply to the blade motor **31**

synchronously with the clock of the first carrier frequency delivered from the first divider **61**.
(92) When the self-travel switch **22** is in ON-state, the second PWM generator **66** generates the second PWM signal to PWM control the electric power supply to the self-travel motor **32** synchronously with the clock of the second carrier frequency delivered from the second divider **62**.
(93) When drive of the blade motor **31** and the self-travel motor **32** is commenced, the first PWM generator **65** and the second PWM generator **66** respectively set duty ratios of the first PWM signal and the second PWM signal to preset minimum values. The duty ratios of the first PWM signal and the second PWM signal are gradually increased at every cycle of the first PWM signal and the second PWM signal until the duty ratios reach respective target duty ratios.
(94) This gradual increase is aimed to perform a so-called soft start, in which the rotational speeds of the blade motor **31** and the self-travel motor **32** are gradually increased after the drive of the motors is commenced. This soft start enables a smooth increase of rotational speeds of the cutting blade **18** and drive wheel **16** after the drive of the motors is commenced, which enables a reduction of uncomfortableness felt by the user due to steep rotational fluctuation when the speeds are increasing.
(95) As shown in FIG. 3, the target duty ratio (DUTY) of the first PWM signal generated in the first PWM generator **65** is fixed at 100%. The blade motor **31** is thus controlled to rotate at its maximum rotational speed by the PWM signal with DUTY at 100%.
(96) Meanwhile, the target duty ratio (DUTY) of the second PWM signal generated in the second PWM generator **66** is set at any duty ratio corresponding to the speed command value received from the speed adjusting dial **24** with the maximum value being 100% as shown in FIG. 3. Accordingly, the rotational speed of the self-travel motor **32** is controlled to be corresponding to the speed command value.
(97) Since the blade motor **31** is a motor that drives the cutting blade **18**, the load applied to the blade motor **31** from the cutting blade **18** when mowing the lawn is greater than the load applied to the self-travel motor **32** from the drive wheel **16** when the lawn mower **10** is traveling. Accordingly, the power consumption of the blade motor **31** is greater than the power consumption of the self-travel motor **32**.
(98) The first switching element **41** that is used to PWM control the electric power supply to the blade motor **31** is therefore likely to generate heat. The first switching element **41** may go into an overheated state and malfunction if the carrier frequency of the PWM control is high.
(99) Accordingly, the carrier frequency of the first PWM signal generated in the first PWM generator **65** is set at a frequency that can protect the first switching element **41** from overheating. Specifically, the first carrier frequency set in the first divider **61** is set at, for example, 500 Hz to protect the first switching element **41** from overheating.
(100) The self-travel motor **32** is required to keep the rotational speed constant to make the lawn mower **10** travel stably. However, if the carrier frequency of a time when PWM controlling the electric power supply to the self-travel motor **32** is set at a low frequency likewise the blade motor **31**, the electric current that flows to the self-travel motor **32** pulsates easily due to the switching operation of the second switching element **42**. This pulsation is a so-called ripple. If the ripples significantly overlap with the electric current, irregularities of torque occur in the self-travel motor **32**, which disables the lawn mower **10** from travelling stably.
(101) The carrier frequency of the second PWM signal generated in the second PWM generator **66** is therefore set at a frequency higher than the first carrier frequency so that the self-travel motor **32** can make the lawn mower **10** travel stably. Specifically, the second carrier frequency set in the second divider **62** is set at, for example, 8 kHz to enable the lawn mower **10** to travel stably.
(102) The motor drive controller **68** then outputs the first PWM signal and the second PWM signal generated as mentioned above in the first PWM generator **65** and the second PWM generator **66** respectively as a drive signal for the first switching element **41** and a drive signal for the second switching element **42**.

(103) As a consequence, the electric power supply to the blade motor **31** and the self-travel motor **32** is PWM controlled respectively by using the first PWM signal and the second PWM signal to respectively drive the blade motor **31** and the self-travel motor **32**.

(104) If the output of the first PWM signal or the second PWM signal is stopped in response to switching of the blade switch **21** or the self-travel switch **22** from ON-state to OFF-state, the motor drive controller **68** delivers a drive signal (hereinafter, brake signal) to the fourth and the fifth switching elements **45**, **46**.

(105) The fourth and the fifth switching elements **45**, **46** are turned to ON-state by the brake signal, deliver a brake current to the blade motor **31** or the self-travel motor **32**, and generate a braking force. Accordingly, when the blade switch **21** or the self-travel switch **22** is turned from ON-state to OFF-state, the blade motor **31** or the self-travel motor **32** can be swiftly stopped.

(106) In the present embodiment, the first divider **61** corresponds to one example of the first carrier frequency setting device in the present disclosure; and the second divider **62** corresponds to one example of the second carrier frequency setting device in the present disclosure. The first PWM generator **65**, the second PWM generator **66**, and the motor drive controller **68** correspond to examples of the controllers in the present disclosure.

(107) <Process>

(108) A control process executed in the control circuit **50** to achieve each of the aforementioned functions will be explained next with reference to the flow charts shown in FIG. **4** to FIG. **9**.

(109) This control process is performed by the CPU executing programs stored in the ROM when the control circuit **50** is in operation in response to the supply of the power supply voltage V_{cc} from the power-supply circuit **38**.

(110) Firstly, after initiation of the control process, elapsed time is measured in **S110** (S: STEP) by counting up a counter for measuring the elapsed time. The elapsed time measured in **S110** is a control-stop time, during which the control to drive or brake the blade motor **31** and the self-travel motor **32** is not executed.

(111) In **S120**, it is determined whether a drive switch of the blade motor **31** or the self-travel motor **32**, more specifically, the blade switch **21** or the self-travel switch **22**, is in ON-state.

(112) If it is determined in **S120** that the drive switch of the blade motor **31** or the self-travel motor **32** is in ON-state, the process proceeds to **S130**. In **S130**, the counter for measuring the elapsed time is cleared, and the process proceeds to **S140**. In **S140**, a motor drive process shown in FIG. **5** is executed, and the process proceeds to **S110**. The motor drive process will be explained in detail later.

(113) If it is determined in **S120** that the drive switches of the blade motor **31** and the self-travel motor **32** are both in OFF-state, the process proceeds to **S150**. In **S150**, it is determined whether the blade motor **31** is currently being braked, more specifically, whether the brake signal is being delivered to the fourth switching element **45**.

(114) If it is determined in **S150** that the blade motor **31** is currently being braked, the process proceeds to **S160**. If it is determined in **S150** that the blade motor **31** is not currently being braked, the process proceeds to **S180**.

(115) In **S160**, the counter for measuring the elapsed time is cleared, and the process proceeds to **S170**. In **S170**, a blade brake control process shown in FIG. **9** is executed, and the process proceeds to **S180**. The blade brake control process will be explained in detail later.

(116) In **S180**, it is determined whether the self-travel motor **32** is currently being braked, more specifically, whether the brake signal is delivered to the fifth switching element **46**.

(117) If it is determined in **S180** that the self-travel motor **32** is currently being braked, the process proceeds to **S190**. If it is determined in **S180** that the self-travel motor **32** is not currently being braked, the process proceeds to **S210**.

(118) In **S190**, the counter for measuring the elapsed time is cleared, and the process proceeds to **S200**. In **S200**, a self-travelling brake control process shown in FIG. **8** is executed, and the process

proceeds to **S210**. The self-travelling brake control process will be explained in detail later.

(119) In **S210**, it is determined whether a counter value of the counter for measuring the elapsed time, which is counted up in **S110**, has reached a preset threshold for determining elapsing of time. If the counter value of the counter for measuring the elapsed time has not reached the threshold, the process proceeds to **S110** and the aforementioned series of processes is executed again.

(120) If it is determined in **S210** that the counter value of the counter for measuring the elapsed time has reached the threshold, the process proceeds to **S220** and the control process is ended by executing a power-off process.

(121) The power-off process is a process to cause the control circuit **50** to stop its operation when the aforementioned counter value reached the threshold, more specifically, when a control-stop time, during which a control to drive or brake the blade motor **31** and the self-travel motor **32** is not executed, has reached a specified waiting time. Thus, in **S220**, the control process is ended by stopping the operation of the power-supply circuit **38**.

(122) The motor drive process executed in **S140** is explained next.

(123) As shown in FIG. 5, in the motor drive process, it is determined first in **S310** whether both of the blade switch **21** and the self-travel switch **22** are in ON-state. If both of the blade switch **21** and the self-travel switch **22** are in ON-state, the process proceeds to **S320** to execute a blade motor control process shown in FIG. 6. After executing the blade motor control process in **S320**, the process proceeds to **S330** to execute a self-travel motor control process shown in FIG. 7, and the motor drive process is ended.

(124) If it is determined in **S310** that both of the blade switch **21** and the self-travel switch **22** are not in ON-state, the process proceeds to **S340** and determines whether the blade switch **21** is in ON-state.

(125) If it is determined in **S340** that the blade switch **21** is in ON-state, the process proceeds to **S350** to determine whether the self-travel motor **32** is currently being braked. If it is determined in **S350** that the self-travel motor **32** is currently being braked, the process proceeds to **S360** to execute the self-travelling brake control process shown in FIG. 8 and proceeds to **S370**. If it is determined in **S350** that the self-travel motor **32** is not currently being braked, the process proceeds to **S370**. In **S370**, the blade motor control process shown in FIG. 6 is executed, and the motor drive process is ended.

(126) If it is determined in **S340** that the blade switch **21** is not in ON-state, in other words, the self-travel switch **22** is in ON-state, the process proceeds to **S380** and determines whether the blade motor **31** is currently being braked.

(127) If it is determined in **S380** that the blade motor **31** is currently being braked, the process proceeds to **S390** to execute the blade brake control process shown in FIG. 9 and proceeds to **S400**. If it is determined in **S380** that the blade motor **32** is not currently being braked, the process proceeds to **S400**. In **S400**, the self-travel motor control process shown in FIG. 7 is executed, and the motor drive process is ended.

(128) The blade motor control process executed in **S320** or **S370** and the self-travel motor control process executed in **S330** or **S400** are explained next.

(129) As shown in FIG. 6, the blade motor control process begins with determining whether the blade motor **31** is currently being braked in **S410**. If it is determined that the blade motor **31** is currently being braked, the process proceeds to **S440** to execute the blade brake control process shown in FIG. 9, and the blade motor control process is ended.

(130) If it is determined in **S410** that the blade motor **31** is not currently being braked, the process proceeds to **S420** to set the duty ratio of the first PWM signal that is to be delivered to the first switching element **41**.

(131) Although the target duty ratio (DUTY) of the first PWM signal is fixed at 100%, in **S420**, the duty ratio is set such that the duty ratio approaches to the target duty ratio every single cycle of the first PWM signal to enable the aforementioned soft start.

(132) In the subsequent **S430**, an output control of the PWM signal is executed, and then the blade motor control process is ended. Specifically, the output control of the PWM signal in this process is a control to generate the first PWM signal, which is pulse-width modulated by using the duty ratio set in **S420**, and to deliver the generated first PWM signal to the first switching element **41**.

(133) The cycle of the first PWM signal is a cycle of the first carrier frequency that is generated by dividing the reference clock in the first divider **61**. In **S430**, the PWM signal with the duty ratio that is set synchronously with this cycle in **S420** is delivered.

(134) As shown in FIG. 7, the self-travel motor control process begins with determining whether the self-travel motor **32** is currently being braked in **S510**. If it is determined that the self-travel motor **32** is currently being braked, the process proceeds to **S540** to execute the self-travelling brake control process shown in FIG. 8, and the self-travel motor control process is ended.

(135) If it is determined in **S510** that the self-travel motor **32** is not currently braked, the process proceeds to **S520** to set the duty ratio of the second PWM signal that is to be delivered to the second switching element **42**.

(136) The target duty ratio (DUTY) of the second PWM signal is set in accordance with the speed command value from the speed adjusting dial **24**. In **S520**, the duty ratio is set such that the duty ratio approaches to the target duty ratio every single cycle of the second PWM signal to enable the aforementioned soft start.

(137) In the subsequent **S530**, an output control of the PWM signal is executed, and then the self-travel motor control process is ended. Specifically, the output control of the PWM signal in this process is a control to generate the second PWM signal, which is pulse-width modulated by using the duty ratio set in **S520**, and to deliver the generated second PWM signal to the second switching element **42**.

(138) The cycle of the second PWM signal is a cycle of the second carrier frequency that is generated by dividing the reference clock in the second divider **62**. In **S530**, the PWM signal with the duty ratio that is set synchronously with this cycle in **S520** is delivered.

(139) The self-travelling brake control process executed in **S200**, **S360**, or **S540** and the blade brake control process executed in **S170**, **S390**, or **S440** will be explained next.

(140) As shown in FIG. 8, the self-travelling brake control process begins with stopping delivery of the second PWM signal to the second switching element **42** to bring the second switching element **42** into OFF-state in **S610**.

(141) In the subsequent **S620**, the brake signal is delivered to the fifth switching element **46** for self-travelling brake to bring the fifth switching element **46** into ON-state. As a consequence, the brake current flows through the self-travel motor **32** which generates the braking force.

(142) In **S630**, output time (hereinafter, self-travelling brake time) of the brake signal towards the fifth switching element **46** is measured by counting up a counter for measuring self-travelling brake time.

(143) In **S640**, it is determined whether the self-travelling brake time measured in **S630** has reached brake time that is preset for determining termination of the self-travelling brake. A time required to stop the travelling of the lawn mower **10** by the self-travelling brake is preset as this brake time.

(144) If it is determined in **S640** that the self-travelling brake time has not reached the brake time for determining termination of the self-travelling brake, the self-travelling brake control process is ended.

(145) If it is determined in **S640** that the self-travelling brake time has reached the brake time for determining termination of the self-travelling brake, the process proceeds to **S650**. In **S650**, delivery of the brake signal to the fifth switching element **46** is stopped, and the brake control of the self-travel motor **32** is ended. In the subsequent **S660**, the counter for measuring the self-travelling brake time is cleared, and the self-travelling brake control process is ended.

(146) As explained above, in the self-travelling brake control process, when the self-travel switch **22** is brought into OFF-state and thus the drive of the self-travel motor **32** is stopped, the braking

force is generated in the self-travel motor **32** for a given length of time which corresponds to the brake time for determining termination of the self-travelling brake. Accordingly, the travelling of the lawn mower **10** can be stopped without fail.

(147) As shown in FIG. **9**, the blade brake control process begins with stopping the delivery of the first PWM signal to the first switching element **41** and bring the first switching element **41** into OFF-state in **S710**.

(148) In the subsequent **S720**, the brake signal is delivered to the fourth switching element **45** for blade brake to bring the fourth switching element **45** into ON-state. As a consequence, brake current flows through the blade motor **31** which generates the braking force.

(149) In **S730**, output time (hereinafter, blade brake time) of the brake signal towards the fourth switching element **45** is measured by counting up a counter for measuring blade brake time.

(150) In **S740**, it is determined whether the blade brake time measured in **S730** has reached a preset brake time TA. If it is determined in **S740** that the blade brake time has not reached the brake time TA, the blade brake control process is ended. A time required for the rotation of the blade motor **31** to be decreased by the blade brake to sufficiently lower the brake current is set as the brake time TA.

(151) If it is determined in **S740** that the blade brake time has reached the brake time TA, the process proceeds to **S750** to determine whether the blade switch **21** is in ON-state.

(152) If it is determined in **S750** that the blade switch **21** is in ON-state, the process proceeds to **S770** and the brake control of the blade motor **31** is ended by terminating the delivery of the brake signal to the fourth switching element **45**. In the subsequent **S780**, the counter for measuring the blade brake time is cleared, and the blade brake control process is ended.

(153) If it is determined in **S750** that the blade switch **21** is not in ON-state, the process proceeds to **S760** to determine whether the blade brake time measured in **S730** has reached a preset brake time TB. A time required for stopping the rotation of the blade motor **31** by using the blade brake is preset as the brake time TB. Accordingly, the brake time TB is longer than the brake time TA.

(154) If it is determined in **S760** that the blade brake time has not reached the brake time TB, the blade brake control process is ended. If it is determined in **S760** that the blade brake time has reached the brake time TB, the brake control of the blade motor **31** is ended in **S770**. In **S780**, the counter for measuring the blade brake time is cleared, and the blade brake control process is ended.

(155) As explained above, in the blade brake control process, when the blade switch **21** is brought into OFF-state and thus the drive of the blade motor **31** is stopped, the brake signal is delivered to the fourth switching element **45** to generate the braking force in the blade motor **31** as shown in FIG. **10**.

(156) If the blade brake time for delivering the brake signal to the fourth switching element **45** has reached the brake time TB, it is determined that the rotation of the blade motor **31** is stopped. Then, the delivery of the brake signal is stopped and the brake control of the blade motor **31** is ended.

(157) If the blade switch **21** is brought into ON-state before the blade brake time has reached the brake time TB, the delivery of the brake signal is stopped with a condition that the blade brake time has reached the brake time TA, which allows recommencement of the drive of the blade motor **31**.

(158) Therefore, if the user stops manipulation of the blade switch **21** when mowing lawns by manipulating the blade switch **21**, he/she can resume mowing lawns by resuming manipulation of the blade switch **21** before the rotation of the blade motor **31** stops.

(159) For example, in the blade brake control process, if the user stops manipulation of the blade switch **21** when mowing lawns by manipulating the blade switch **21**, the braking force may be applied to the blade motor **31** until the rotation of the blade motor **31** stops. However, such configuration decreases usability of the lawn mower **10** for a user who wants to mow lawns by resuming the drive of the blade motor **31** in a state where the blade motor **31** is not yet stopped.

(160) Meanwhile, in the present embodiment, the user can resume the drive of the blade motor **21** by manipulating the blade switch **21** even when the blade motor **31** is not stopped by the brake

control if the brake control is executed for the brake time TA or longer. Therefore, the lawn mower **10** of the present embodiment enables efficient lawn mowing and improvement of usability for the user.

(161) <Effect>

(162) As it has been explained, in the lawn mower **10** of the present embodiment, the electric power supply to the blade motor **31** and the self-travel motor **32** is PWM controlled respectively by the first PWM signal and the second PWM signal generated respectively by the first carrier frequency and the second carrier frequency, whose frequencies are different from each other.

(163) Accordingly, even though the drive of the blade motor **31** and the self-travel motor **32** are controlled by the common control circuit **50**, the carrier frequency of a time when PWM controlling the electric power supply to these motors can be set individually for each motor. Therefore, the electric power supply to the blade motor **31** and the self-travel motor **32** can be PWM controlled individually at the carrier frequency suitable for each motor.

(164) In other words, since the load applied to the blade motor **31** from the cutting blade **18** is large, the electric power supply to the blade motor **31** is greater than the electric power supply to the self-travel motor **32**; and therefore, the electric current that flows through the first switching element **41** is greater than the electric current that flows through the second switching element **42**.

(165) Meanwhile, in the present embodiment, the first carrier frequency of a time when PWM controlling the first switching element **41** can be set at a frequency lower than the second carrier frequency. This enables a reduction of switching losses in the first switching element **41**, which can inhibit overheating of the first switching element **41**.

(166) Furthermore, since it is not necessary to increase the dielectric withstanding voltage of the first switching element **41** to protect the first switching element **41** from malfunctions caused by overheating, it is possible to use an element that has a low dielectric withstanding voltage but is inexpensive as the first switching element **41**. This enables a reduction of the cost of the lawn mower **10**.

(167) The second carrier frequency can be set higher than the first carrier frequency to be able to reduce the torque fluctuation of the self-travel motor **32**. Accordingly, the lawn mower **10** of the present embodiment can stably travel at a desired speed.

(168) In addition, in the lawn mower **10** of the present embodiment, the first and the second switching elements **41**, **42**, which are used to PWM control the electric power supply to the blade motor **31** and the self-travel motor **32**, are configured such that their turn-off time is longer than the turn-on time.

(169) Accordingly, it is possible to protect the first and the second switching elements **41**, **42** from a high voltage by reducing the high-value peak voltage generated when turning off the first and the second switching elements **41**, **42**.

Other Embodiments

(170) Although the embodiment of the present disclosure has been explained above, the present disclosure may be implemented in various forms without being limited to the aforementioned embodiment.

(171) In the aforementioned embodiment, it is explained that the first carrier frequency of a time when PWM controlling the electric power supply to the blade motor **31** is 500 Hz. Nevertheless, the first carrier frequency may be set within a range between 100 Hz and 1 kHz.

(172) In the aforementioned embodiment, the second carrier frequency of a time when PWM controlling the electric power supply to the self-travel motor **32** is 8 kHz. Nevertheless, the second carrier frequency may be set within a range between 8 kHz and 20 kHz.

(173) In other words, if the carrier frequency is set within a frequency region between 1 kHz and 8 kHz when PWM controlling the electric power supply to the motors, the switching noise generated by turning on and of the switches may be heard by a person in the surroundings such as the user.

(174) However, if the first carrier frequency and the second carrier frequency are set within the

aforementioned frequency ranges, it is possible to inhibit the switching noise generated when PWM controlling the electric power supply to the motors from being heard by the user and the like, which enables a reduction of uncomfortableness felt by the user.

(175) Particularly, the first carrier frequency may be set within a range between 300 Hz and 700 Hz; and the second carrier frequency may be set within a range between 8 kHz and 10 kHz.

(176) In the aforementioned embodiment, it is explained that the first carrier frequency and the second carrier frequency are set by dividing the reference clock in the control circuit **50**. Nevertheless, the first carrier frequency and the second carrier frequency may also be set by multiplying the reference clock.

(177) Alternatively, an oscillator that generates a clock signal of the first carrier frequency or the second carrier frequency may be provided; and by dividing or multiplying an output from the oscillator, a clock signal having a carrier frequency different from thus generated clock signal may be generated.

(178) In the aforementioned embodiment, the lawn mower **10** is described. Nevertheless, the configuration of the aforementioned embodiment can be applied to electric work machines that include two or more motors for different uses, for example, a reinforcing steel binding machine having a motor for binding reinforcing steels and a motor for transferring wires for binding. In addition, the configuration of the aforementioned embodiment can also be applied to self-travelling work machines, for example, a hand cart, a cultivator, and a floor cleaning machine.

(179) In the aforementioned embodiment, examples are provided where the blade motor **31** and the self-travel motor **32** are both brushed motors. Nevertheless, one or both of the blade motor **31** and the self-travel motor **32** may be brushless motors. In other words, the two motors of the present disclosure may be any motors as long as the electric power supply thereto can be PWM controlled.

(180) Two or more functions of one element in the aforementioned embodiments may be achieved by two or more elements, and one function of one element may be achieved by two or more elements. In addition, two or more functions of two or more elements may be achieved by one element, and one function of two or more elements may be achieved by one element. A part of the configurations in the aforementioned embodiments may be omitted. Furthermore, at least a part of the configurations of the aforementioned embodiments may be added to or replaced with another configuration of the aforementioned embodiments.

(181) Other than the electric work machines, the present disclosure may be embodied in various forms, for example, a system that includes the electric work machine as an element; a program that makes a computer function as the electric work machine; a non-transitory tangible recording medium, such as a semiconductor memory, that stores the program; and a control method.

Claims

1. An electric work machine comprising: a first motor configured to drive a first drive object; a second motor configured to drive a second drive object; a first current conduction path configured to couple a direct-current power source to the first motor; a second current conduction path configured to couple the direct-current power source to the second motor; a first switch disposed on the first current conduction path; a second switch disposed on the second current conduction path; a first carrier frequency setting device configured to set a first carrier frequency at a time when a first PWM signal controls electric power supplied to the first motor by turning on and off the first switch; a second carrier frequency setting device configured to set a second carrier frequency at a time when a second PWM signal controls electric power supplied to the second motor by turning on and off the second switch such that the second carrier frequency is different from the first carrier frequency; and, a controller configured to control electric power supplied to the first motor by generating the first PWM signal at the first carrier frequency and delivering the first PWM signal to the first switch in response to an input of a first motor drive command, and control electric power

- supplied to the second motor by generating the second PWM signal at the second carrier frequency and delivering the second PWM signal to the second switch in response to an input of a second motor drive command, wherein the electric work machine comprises a lawn mower.
2. The electric work machine according to claim 1, wherein the first carrier frequency is set lower than the second carrier frequency.
 3. The electric work machine according to claim 2, wherein the first drive object is a blade, and wherein the first motor is configured to rotate the blade.
 4. The electric work machine according to claim 3, wherein the second drive object is a wheel that enables the electric work machine to travel, and wherein the second motor is configured to rotate the wheel.
 5. The electric work machine according to claim 4, wherein the first carrier frequency is set within a range between 100 Hz and 1 kHz.
 6. The electric work machine according to claim 5, wherein the first carrier frequency is set within a range between 300 Hz and 700 Hz.
 7. The electric work machine according to claim 5, wherein the second carrier frequency is set within a range between 8 kHz and 20 kHz.
 8. The electric work machine according to claim 7, wherein the second carrier frequency is set within a range between 8 kHz and 10 kHz.
 9. The electric work machine according to claim 1, wherein the first switch and/or the second switch is configured such that a turn-off time, during which a state thereof changes from ON-state to OFF-state by the first PWM signal and/or the second PWM signal, is longer than a turn-on time, during which the state changes from the OFF-state to the ON-state.
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